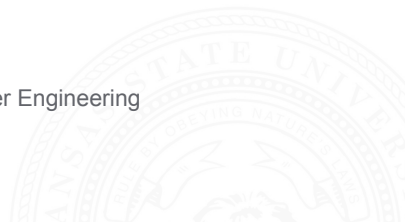


Single-Phase AC Circuits I

From a time waveform to an RMS phasor and back

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Lecture outline

- 1 read amplitude, frequency, period, and phase from a cosine waveform
- 2 convert peak values to RMS values and explain what RMS means
- 3 convert among time-domain, polar, and rectangular forms, and
- 4 choose the right form, validate compatibility, and check a phasor sum.

A phasor is compact when frequency and the RMS convention are declared first.

Starting problem

STARTING PROBLEM

For

$$v(t) = 120\sqrt{2} \cos(377t + 30^\circ) \text{ V},$$

**what are its peak value, RMS value,
frequency, and phase angle?**

Starting-problem solution

Peak

$$\hat{V} = 169.7 \text{ V}$$

Coefficient: $120\sqrt{2} \text{ V}$.

RMS

$$V_{\text{rms}} = 120 \text{ V}$$

Peak divided by $\sqrt{2}$.

Frequency

$$f = 60.0 \text{ Hz}$$

From $377/(2\pi)$; radians per second to cycles per second.

Phase

$$\phi = +30^\circ$$

The waveform leads a cosine reference.

Peak and RMS describe size. Frequency describes repetition. Phase describes relative position.

Sinusoid

$$x(t) = \hat{X} \cos(\omega t + \phi)$$

Amplitude

 $\hat{X}$

Peak distance from zero. Same unit as x .

Phase angle

 ϕ

Horizontal position relative to the reference.

Angular frequency

 ω

Radians per second.

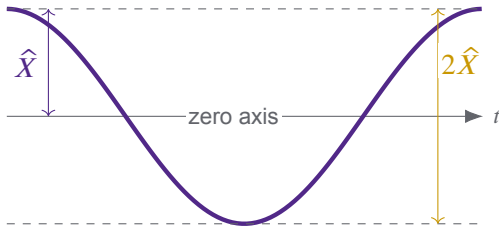
Instantaneous value

 $x(t)$

Value at one selected time.

Changing $\hat{X}$ stretches vertically. Changing ω compresses in time. Changing ϕ shifts horizontally.

Waveform magnitudes



Peak

$$\hat{X}$$

Zero to maximum.

Peak-to-peak

$$2\hat{X}$$

Minimum to maximum.

Sinusoidal RMS

$$X_{\text{rms}} = \hat{X}/\sqrt{2}$$

Equivalent heating value.

The coefficient of a time-domain cosine normally gives the peak value. Divide by $\sqrt{2}$ to obtain RMS.

Frequency and period

$$T = \frac{1}{f}, \quad \omega = 2\pi f, \quad T = \frac{2\pi}{\omega}.$$

At 60 Hz

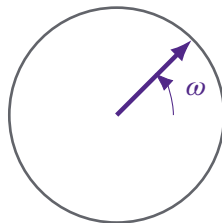
$$T = 16.67 \text{ ms}$$

One cycle every 0.01667 s.

Over one period

$$\omega T = 2\pi$$

The angle advances by one complete cycle.

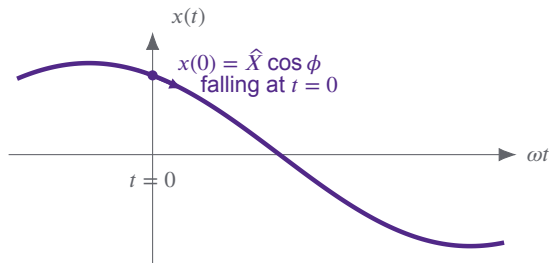


2π radians per cycle

Hertz counts cycles per second. Radians per second tracks angular motion.

What phase means

A phase angle requires a cosine reference and a declared time origin, $t = 0$



$$\theta(t) = \omega t + \phi, \quad \theta(0) = \phi.$$

Instantaneous phase

$\theta(t)$

Changes as time advances.

Fixed phase offset

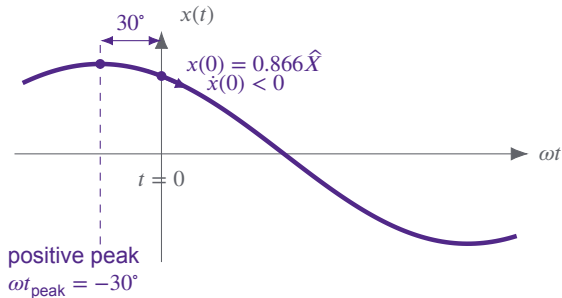
ϕ

Constant; equal to $\theta(0)$.

Without a declared $t = 0$, one isolated curve has no absolute phase angle; only relative displacement can be read.

Read phase from the time curve

At $t = 0$, the value gives possible angles and the curve direction selects the sign



$$\frac{x(0)}{\hat{X}} = 0.866 \Rightarrow \phi = \pm 30^\circ.$$

$$\dot{x}(0) = -\omega \hat{X} \sin \phi < 0, \\ \Rightarrow \sin \phi > 0 \Rightarrow \phi = +30^\circ.$$

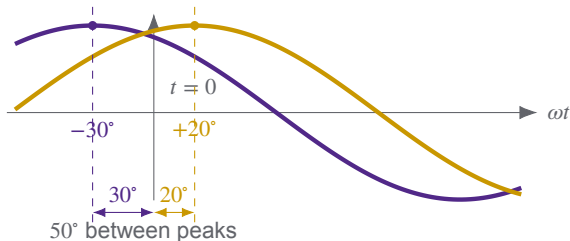
Equivalent peak check

$$\omega t_{\text{peak}} + \phi = 0 \Rightarrow \phi = -\omega t_{\text{peak}}.$$

At $t = 0$, value gives candidate angles and slope selects the sign; the peak method solves the same equation at a positive peak.

Phase difference

Use the same $t = 0$ axis to find each phase, then subtract



$$\phi_1 = -(-30^\circ) = +30^\circ,$$

$$\phi_2 = -(+20^\circ) = -20^\circ.$$

$$\Delta\phi = \phi_1 - \phi_2 = 50^\circ.$$

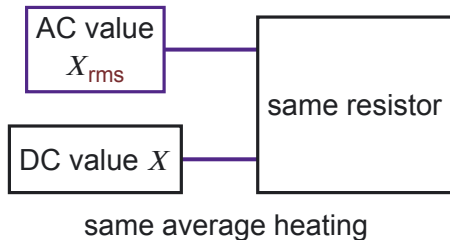
Purple is left/earlier, so it leads gold by 50° ; equivalently, gold lags purple by 50° .

Compare corresponding points on the original curves: left = earlier = leads; right = later = lags.

RMS

$$X_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt}.$$

- 1 square the waveform so positive and negative halves both contribute
- 2 average the squared value over one period
- 3 take the square root to recover the original unit.



RMS is the square root of the mean square over one period; physically, it is the DC value that produces the same average heating in a resistor.

RMS derivation

For $x(t) = \hat{X} \cos(\omega t + \phi)$, use $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$:

$$\begin{aligned} X_{\text{rms}}^2 &= \frac{\hat{X}^2}{T} \int_0^T \cos^2(\omega t + \phi) dt \\ &= \frac{\hat{X}^2}{2T} \int_0^T [1 + \cos(2\omega t + 2\phi)] dt \\ &= \frac{\hat{X}^2}{2T} \left[t + \frac{\sin(2\omega t + 2\phi)}{2\omega} \right]_0^T \\ &= \frac{\hat{X}^2}{2T} \left[T + \frac{\sin(2\omega T + 2\phi) - \sin(2\phi)}{2\omega} \right]. \end{aligned}$$

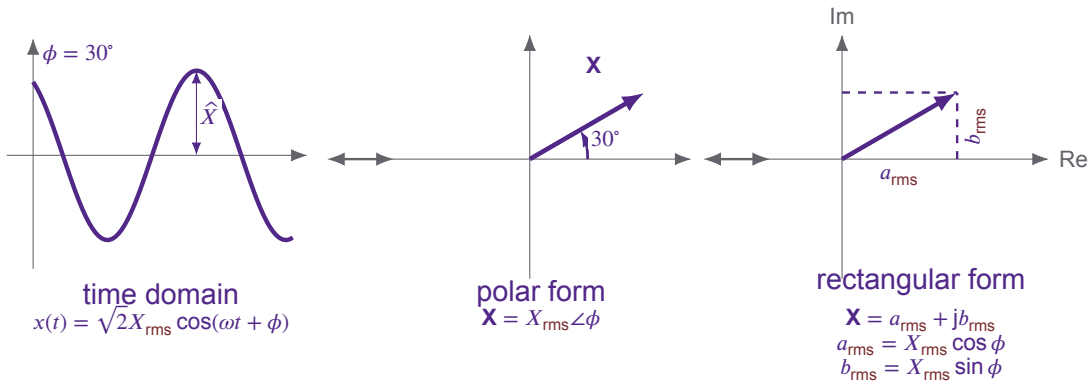
$$T = \frac{2\pi}{\omega} \implies \omega T = 2\pi \implies \sin(2\omega T + 2\phi) = \sin(2\phi),$$

$$X_{\text{rms}}^2 = \frac{\hat{X}^2}{2},$$

$$X_{\text{rms}} = \frac{\hat{X}}{\sqrt{2}}.$$

Three equivalent representations

Time domain, polar form, and rectangular form



Time domain to polar form

$$x(t) = \sqrt{2}X_{\text{rms}} \cos(\omega t + \phi) \iff \mathbf{X} = X_{\text{rms}} \angle \phi$$

Kept in the phasor

- phasor magnitude X_{rms} (the waveform's RMS value).
- phase angle ϕ .
- the physical unit.

Declared once for the circuit

- common angular frequency ω .
- cosine reference.
- all phasor values use the RMS convention.

Convert a peak coefficient to an RMS phasor by dividing the magnitude by $\sqrt{2}$.

Polar form to time domain

If the common angular frequency is known,

$$\mathbf{X} = X_{\text{rms}} \angle \phi \implies x(t) = \sqrt{2} X_{\text{rms}} \cos(\omega t + \phi).$$

1 Restore the peak

$$\hat{X} = \sqrt{2} X_{\text{rms}}.$$

2 Restore the common rate

Use the declared ω .

3 Keep the phasor angle

Use the same ϕ in the cosine.

4 Restore the unit

A voltage phasor returns to volts.

Project lens: PowerWorld voltage magnitudes and angles describe RMS phasors. Instantaneous samples remain in the time-domain waveform.

Polar form to rectangular form

$$\mathbf{X} = X_{\text{rms}} \angle \phi = X_{\text{rms}} (\cos \phi + j \sin \phi) = a_{\text{rms}} + j b_{\text{rms}}$$

Resolve the two components

$$a_{\text{rms}} = X_{\text{rms}} \cos \phi$$

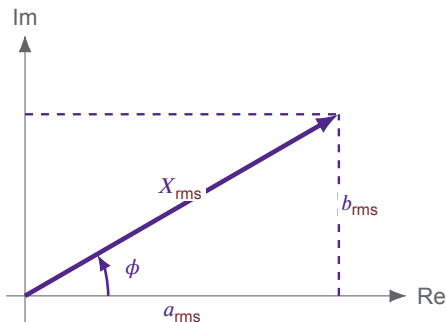
$$b_{\text{rms}} = X_{\text{rms}} \sin \phi$$

Check the signs before calculating

- a_{rms} is the horizontal, or real, RMS component.
- b_{rms} is the vertical, or imaginary, RMS component.
- the quadrant of ϕ predicts the signs of both components.

Polar and rectangular forms describe the same RMS phasor; changing coordinates introduces no $\sqrt{2}$.

Rectangular form to polar form



$$\mathbf{X} = a_{\text{rms}} + jb_{\text{rms}} = X_{\text{rms}} \angle \phi$$

$$X_{\text{rms}} = \sqrt{a_{\text{rms}}^2 + b_{\text{rms}}^2}$$

$$\phi = \text{atan2}(b_{\text{rms}}, a_{\text{rms}})$$

Quadrant check

The signs of a_{rms} and b_{rms} must agree with the quadrant of ϕ .

Use $\text{atan2}(b_{\text{rms}}, a_{\text{rms}})$ so that the angle is placed in the correct quadrant.

Choose polar or rectangular form

Use polar form

$$X_{\text{rms}} \angle \alpha \times Y_{\text{rms}} \angle \beta = (X_{\text{rms}} Y_{\text{rms}}) \angle (\alpha + \beta)$$

$$\frac{X_{\text{rms}} \angle \alpha}{Y_{\text{rms}} \angle \beta} = \frac{X_{\text{rms}}}{Y_{\text{rms}}} \angle (\alpha - \beta)$$

Use rectangular form

$$\begin{aligned} (a_{\text{rms}} + jb_{\text{rms}}) + (c_{\text{rms}} + jd_{\text{rms}}) \\ = (a_{\text{rms}} + c_{\text{rms}}) + j(b_{\text{rms}} + d_{\text{rms}}) \end{aligned}$$

$$\begin{aligned} (a_{\text{rms}} + jb_{\text{rms}}) - (c_{\text{rms}} + jd_{\text{rms}}) \\ = (a_{\text{rms}} - c_{\text{rms}}) + j(b_{\text{rms}} - d_{\text{rms}}) \end{aligned}$$

Multiply and divide in polar form. Add and subtract in rectangular form.

Before adding, confirm compatibility

All three conditions must match

- 1 the same angular frequency ω
- 2 the same physical quantity and unit
- 3 the same phasor convention and angle reference

Examples

- $\mathbf{V}_1 + \mathbf{V}_2$ is valid when both are voltage phasors at the same ω .
- $\mathbf{V} + \mathbf{I}$ is not an addition of like physical quantities.
- different frequencies cannot be combined into one common-frequency phasor sum.

Compatibility is checked before choosing coordinates or doing arithmetic.

Phasor addition is component addition

1 Convert both phasors to components

$$\mathbf{X} = X_{\text{rms}} \angle \alpha = X_{\text{rms}} \cos \alpha + jX_{\text{rms}} \sin \alpha \text{ and } \mathbf{Y} = Y_{\text{rms}} \angle \beta = Y_{\text{rms}} \cos \beta + jY_{\text{rms}} \sin \beta.$$

2 Add matching components

$$a_{Z,\text{rms}} = X_{\text{rms}} \cos \alpha + Y_{\text{rms}} \cos \beta,$$

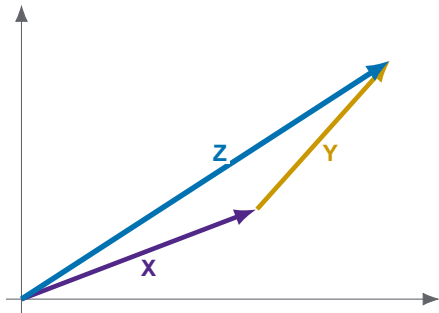
$$b_{Z,\text{rms}} = X_{\text{rms}} \sin \alpha + Y_{\text{rms}} \sin \beta, \quad \mathbf{Z} = a_{Z,\text{rms}} + jb_{Z,\text{rms}}.$$

3 Convert the result back to polar form

$$Z_{\text{rms}} = \sqrt{a_{Z,\text{rms}}^2 + b_{Z,\text{rms}}^2}, \theta = \text{atan2}(b_{Z,\text{rms}}, a_{Z,\text{rms}}), \text{ and } \mathbf{Z} = Z_{\text{rms}} \angle \theta.$$

Do not add polar magnitudes and angles separately; add the real and imaginary components.

Phasor addition as vectors



1 Place vectors head to tail

Move **Y** without changing its length or direction.

2 Draw the resultant

Connect the starting point to the final tip.

$$\mathbf{Z} = \mathbf{X} + \mathbf{Y}$$

The diagram shows the direction of the sum; rectangular components give its numerical value.

SECTION

Worked example phasor conversion

Convert, compare, add, and reconstruct one coherent set of sinusoidal quantities.



phasor conversion setup

Time-domain data

$$v_s(t) = 120\sqrt{2} \cos(377t + 30^\circ) \text{ V}$$

$$i(t) = 10\sqrt{2} \cos(377t - 20^\circ) \text{ A}$$

Additional voltage phasor

$$\mathbf{V}_x = V_{x,\text{rms}} \angle(-60^\circ) = 40 \angle(-60^\circ) \text{ V}.$$

Find

- 1 $\mathbf{V}_s$ and $\mathbf{I}$
- 2 the relative voltage-current angle
- 3 $\mathbf{V}_T = \mathbf{V}_s + \mathbf{V}_x$, and
- 4 $v_T(t)$ in cosine form.

phasor conversion Step 1

1 Identify the peak coefficient

$$\hat{V}_s = 120\sqrt{2} \text{ V.}$$

2 Divide by root two

$$V_{s,\text{rms}} = \frac{120\sqrt{2}}{\sqrt{2}} = 120 \text{ V.}$$

3 Keep the cosine phase

$$\mathbf{V}_s = V_{s,\text{rms}} \angle 30^\circ = 120 \angle 30^\circ \text{ V.}$$

The peak-to-RMS conversion changes the magnitude by $\sqrt{2}$ and preserves the $+30^\circ$ phase.

phasor conversion Step 2

1 Identify the peak coefficient

$$\hat{I} = 10\sqrt{2} \text{ A.}$$

2 Divide by root two

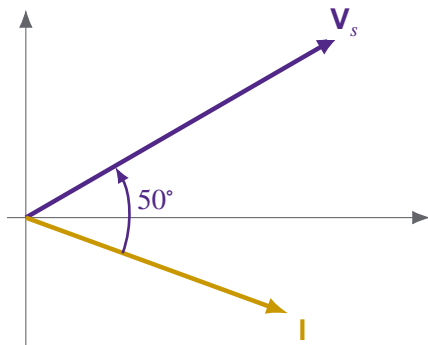
$$I_{\text{rms}} = \frac{10\sqrt{2}}{\sqrt{2}} = 10 \text{ A.}$$

3 Keep the cosine phase

$$\mathbf{I} = I_{\text{rms}} \angle (-20^\circ) = 10 \angle (-20^\circ) \text{ A.}$$

A negative current angle places the current clockwise from the 0° reference.

phasor conversion Step 3



$$\theta_V - \theta_I = 30^\circ - (-20^\circ) = \boxed{50^\circ}.$$

Voltage

leads current

Its angle is more counterclockwise.

Current

lags voltage

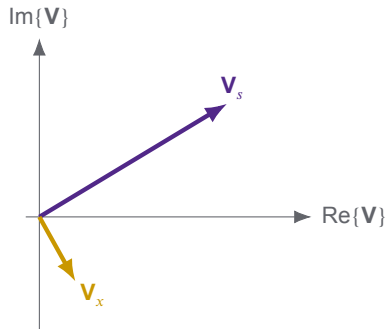
Its angle is 50° smaller.

Subtract angles in a declared order, then attach the words lead or lag.

phasor conversion Step 4

$$\begin{aligned}\mathbf{V}_s &= V_{s,\text{rms}}(\cos 30^\circ + j \sin 30^\circ) \\ &= 120(\cos 30^\circ + j \sin 30^\circ) \\ &= 103.923 + j60.000 \text{ V},\end{aligned}$$

$$\begin{aligned}\mathbf{V}_x &= V_{x,\text{rms}}(\cos(-60^\circ) + j \sin(-60^\circ)) \\ &= 40(\cos(-60^\circ) + j \sin(-60^\circ)) \\ &= 20.000 - j34.641 \text{ V}.\end{aligned}$$



phasor conversion Step 5

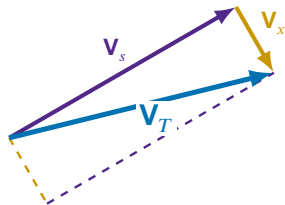
$$\mathbf{V}_T = \mathbf{V}_s + \mathbf{V}_x$$

$$a_{T,\text{rms}} = 103.923 + 20.000 = 123.923$$

$$b_{T,\text{rms}} = 60.000 - 34.641 = 25.359$$

$$\mathbf{V}_T = a_{T,\text{rms}} + jb_{T,\text{rms}}$$

$$\boxed{\mathbf{V}_T = 123.923 + j25.359 \text{ V}}$$



Convert both phasors to rectangular form, then add real and imaginary components.

phasor conversion Step 6

$$V_{T,\text{rms}} = |\mathbf{V}_T| = \sqrt{123.923^2 + 25.359^2} = 126.491 \text{ V}$$

$$\angle \mathbf{V}_T = \text{atan2}(25.359, 123.923) = 11.565^\circ$$

$$\begin{aligned} v_T(t) &= \sqrt{2} V_{T,\text{rms}} \cos(377t + 11.565^\circ) \\ &= \boxed{126.491 \sqrt{2} \cos(377t + 11.565^\circ) \text{ V}}. \end{aligned}$$

Reverse-conversion check

$$\begin{aligned} \hat{V}_T &= \sqrt{2} V_{T,\text{rms}} \\ &= \sqrt{2}(126.491) = 178.885 \text{ V} \end{aligned}$$

$$\begin{aligned} a_{T,\text{rms}} &= V_{T,\text{rms}} \cos(11.565^\circ) \\ &= 123.923 \text{ V} \end{aligned}$$

$$\begin{aligned} b_{T,\text{rms}} &= V_{T,\text{rms}} \sin(11.565^\circ) \\ &= 25.359 \text{ V} \end{aligned}$$

Practice problem

$$v_a(t) = 80\sqrt{2} \cos(377t - 25^\circ) \text{ V}$$

$$v_b(t) = 60\sqrt{2} \cos(377t - 70^\circ) \text{ V}$$

Find

The RMS phasor $\mathbf{V}_c = \mathbf{V}_a + \mathbf{V}_b$ and the corresponding time waveform $v_c(t)$.

Both waveforms use the 60-Hz system frequency and the same cosine reference.

Rectangular addition

1 First voltage

$$\begin{aligned}\mathbf{V}_a &= V_{a,\text{rms}}\angle(-25^\circ) \\ &= 80\angle(-25^\circ) = 72.505 - j33.809 \text{ V}.\end{aligned}$$

2 Second voltage

$$\begin{aligned}\mathbf{V}_b &= V_{b,\text{rms}}\angle(-70^\circ) \\ &= 60\angle(-70^\circ) = 20.521 - j56.382 \text{ V}.\end{aligned}$$

3 Add rectangular components

$$\mathbf{V}_c = a_{c,\text{rms}} + jb_{c,\text{rms}},$$

$$\mathbf{V}_c = 93.026 - j90.192 \text{ V}.$$

Both imaginary parts are negative, so the sum must remain below the real axis.

Polar reconstruction

$$\begin{aligned}\mathbf{V}_c &= V_{c,\text{rms}} \angle (-44.114^\circ) \\ &= 129.569 \angle (-44.114^\circ) \text{ V}\end{aligned}$$

$$\begin{aligned}v_c(t) &= \sqrt{2} V_{c,\text{rms}} \cos(377t - 44.114^\circ) \\ &= \boxed{129.569 \sqrt{2} \cos(377t - 44.114^\circ) \text{ V}}.\end{aligned}$$

Reverse-conversion check

$$\begin{aligned}a_{c,\text{rms}} &= V_{c,\text{rms}} \cos(-44.114^\circ) \\ &= 93.026 \text{ V}\end{aligned}$$

$$\begin{aligned}b_{c,\text{rms}} &= V_{c,\text{rms}} \sin(-44.114^\circ) \\ &= -90.192 \text{ V}\end{aligned}$$

The reverse conversion reproduces both rectangular components.

RMS waveform reconstruction

$\mathbf{X} = X_{\text{rms}} \angle (-40^\circ) = 50 \angle (-40^\circ)$ **at** $\omega = 377$ rad/s.

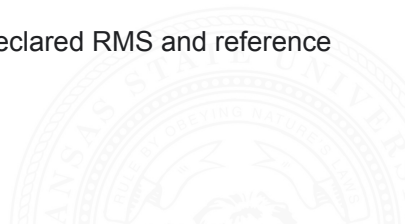
Time waveform

$$\begin{aligned} x(t) &= \sqrt{2} X_{\text{rms}} \cos(377t - 40^\circ) \\ &= 50\sqrt{2} \cos(377t - 40^\circ) \end{aligned}$$

Three checks

peak = $50\sqrt{2}$, frequency = 60.0 Hz,
and phase = -40° .

Project lens: A bus-voltage magnitude and angle require declared RMS and reference conventions.



Phasor summary

Knowledge to carry forward

- **Representation:** one waveform; one phasor in polar or rectangular coordinates.
- **Magnitude:** waveform coefficients are peak values; phasor magnitudes are RMS values.
- **Conversion:** use cosine and sine for components; use magnitude and atan2 for the inverse.
- **Compatibility:** match ω , physical quantity and unit, convention, and angle reference.
- **Operation:** add or subtract in rectangular form; multiply or divide in polar form.
- **Check and reconstruct:** verify signs and reverse conversion; then restore peak, ω , phase, and unit.

Declare conventions, convert, choose the form, compute, check, and reconstruct.